

Math Workshop for Political Science

University of Florida

Syllabus: Autumn 2024

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Class location: 101 Anderson Hall
Class time: Various
Office hours: By Appointment

Description

The purpose of this workshop is to provide incoming first year Ph.D. students with some fundamental skills in various mathematical techniques that are used in political science, regardless of sub-specialty, and generally to prepare students for the first-year methods sequence. The workshop is also open to continuing students who feel that they would gain from participating in the course.

It's important to keep a few things in mind. First, I am not a mathematician. I took one semester of community college calculus that I immediately forgot. The bar here is very low. Second, we will focus on covering the things necessary for you to understand quantitative methods in political science. Some of this stuff might seem esoteric, but I promise it directly ties to something we will cover during the first two methods courses. Third, this course is not graded and your performance is not a reflection of your self-worth.

NB: This course is typically 10 days. We only have 7, so we will have to move fast and cut out some topics. We maybe have to adjust the schedule somewhat.

Textbook

Moore, W.H. and Siegel, D.A., 2013. *A Mathematics Course for Political and Social Research*. Princeton University Press. (SM)

This is a good textbook for getting started with R programming, a topic we won't have time to cover: Wickham, H. and Grolemund, G., 2017. *R for Data Science*. O'Reilly. (WG)

General resources

- Another popular, but more technical, textbook is: Simon, Carl P. and Lawrence Blume., 1994, *Mathematics for Economists*. Vol. 7. New York: Norton.
- MIT Open Courseware (Mathematics): <http://ocw.mit.edu/courses/#mathematics>
- Khan Academy: <http://www.khanacademy.org/>

- Brightstorm: <http://www.brightstorm.com/math/>
- MathTV: http://www.mathtv.com/videos_by_topic

Class Format

The class will meet in-person for an extensive lecture, followed by students completing problem sets after class on their own. For each module, students will be expected to do the textbook readings, consult online videos, and complete a problem set. During each class, we will review the problem set from the previous day's material and then I will answer questions about the current day's material. For example, on day 2 we will review day 1's problem set and answer clarifying questions about day 2's topics.

R Introduction

The compressed nature of this class makes it impossible to give students a comprehensive introduction to all of the tools they will need in the first year methods sequence. However, I am available to provide a brief R and computational social science introduction to interested students.

Class schedule

Day 1 (Monday, August 12; 9:00 am–11:30 am): Introduction, Pre-test, Notation and Definitions, and Some Basic Mathematics

- **Reading** (don't panic): SM (3–21, Ch.2, 44–73)
- Definition of a variable and real number systems
- Set notation and relationships
- Definition of independent and dependent variables
- Discussion of interval notation
- Definitions of types of functions
- Commutative, associative, and distributive laws
- Concepts of inequality and absolute value
- Exponent rules
- Summation and product operators
- Factorials, permutations, and combinations
- Solving equations, inequalities, and for roots
 - Single and multiple variables
 - Quadratic formula
 - Factoring

- Logarithms and rules

Day 2 (Tuesday, August 12; 1:00 pm–3:30 pm): Linear Algebra I

- **Reading:** SM (275–288, 297–298)
- Linear equations and linear systems
- Method of elimination
- Definition of matrices and vectors
- Matrix operators
- Transposes
- Dot product and matrix multiplication
- Matrix representation of systems of equations

Day 3 (Wednesday, August 14; 9:00 am–11:30 am): Linear Algebra II

- **Reading:** SM (289–297, 304–315)
- Linear dependence/independence
- Properties of matrix operators
- Definition of identity, zero, and idempotent matrices
- Reduced row/row echelon form and solving linear systems of equations Gauss-Jordan Reduction/Elimination

Day 4 (Thursday, August 15, 3:00–5:30 pm): Linear Algebra III

- **Reading:** SM (289–293, 309, 315–324)
- Inverses
- Conditions for nonsingularity of matrix
- Definition of matrix rank
- Determinants
- Matrix inversion
- Trace of a matrix

Day 5 (Monday, August 19, 9:00 am–11:30 am): Calculus I

- **Reading:** SM (81–92, Ch.5)
- Limits
- The difference quotient
- The derivative
- Rules of differentiation for a function of one variable
- Rules of differentiation involving two or more functions of the same variable

- Derivative of exponential and log functions
- Partial differentiation
- Second and higher derivatives

Day 6 (Tuesday, August 20, 1:30–4:00 pm): Calculus II

- **Reading:** SM (Ch. 8)
- Definition of optimum and extreme values
- Relative maximum and minimum
- Second-derivative test
- Constrained optimization and Lagrange Multipliers (maybe)

Day 7 (Wednesday, August 21, 1:30–4:00 pm): Calculus III

- **Reading:** SM (133-143)
- Antidifferentiation
- Areas and Riemann sums
- Indefinite and definite integrals
- Fundamental Theorem of Calculus